

Brainstorming & Idea Prioritization Template

Date	04 July 2026
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Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Raman Harshitha	Smart Crop Recommendation using N, P, K, temperature, humidity, pH, rainfall	ML / Prediction	G1
2	Raman Harshitha	Plant Disease Detection from symptoms and environmental clues	ML / Classification	G2
3	Raman Harshitha	Image-based plant disease identification using CNN (leaf photo upload)	Deep Learning / CV	G2
4	Raman Harshitha	Fertilizer recommendation based on crop and NPK soil status	ML / Advisory	G1
5	Raman Harshitha	Real-time weather integration using free API (Open-Meteo) for auto-fill	API / Integration	G3
6	Raman Harshitha	Agricultural research and policy analytics dashboard with charts and CSV export	Analytics / BI	G4

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
G1	Crop Recommendation + Fertilizer Advisor	High	High	1st	Yes
G2	Plant Disease Detection (symptom + image)	High	High	2nd	Yes
G3	Real-time Weather Auto-fill (Open-Meteo API)	High	Medium	3rd	Yes
G4	Agricultural Research & Policy Analytics Dashboard	Medium	Medium	4th	Yes

Selected Concept: OptiCrop — a pure-Python local web application (no framework dependency) combining ML-based crop recommendation (Random Forest, Decision Tree, SVC ensemble), symptom-based and image-based plant disease identification, fertilizer guidance, real-time weather via Open-Meteo API, and an interactive research analytics dashboard with Chart.js visualisations and CSV export.